

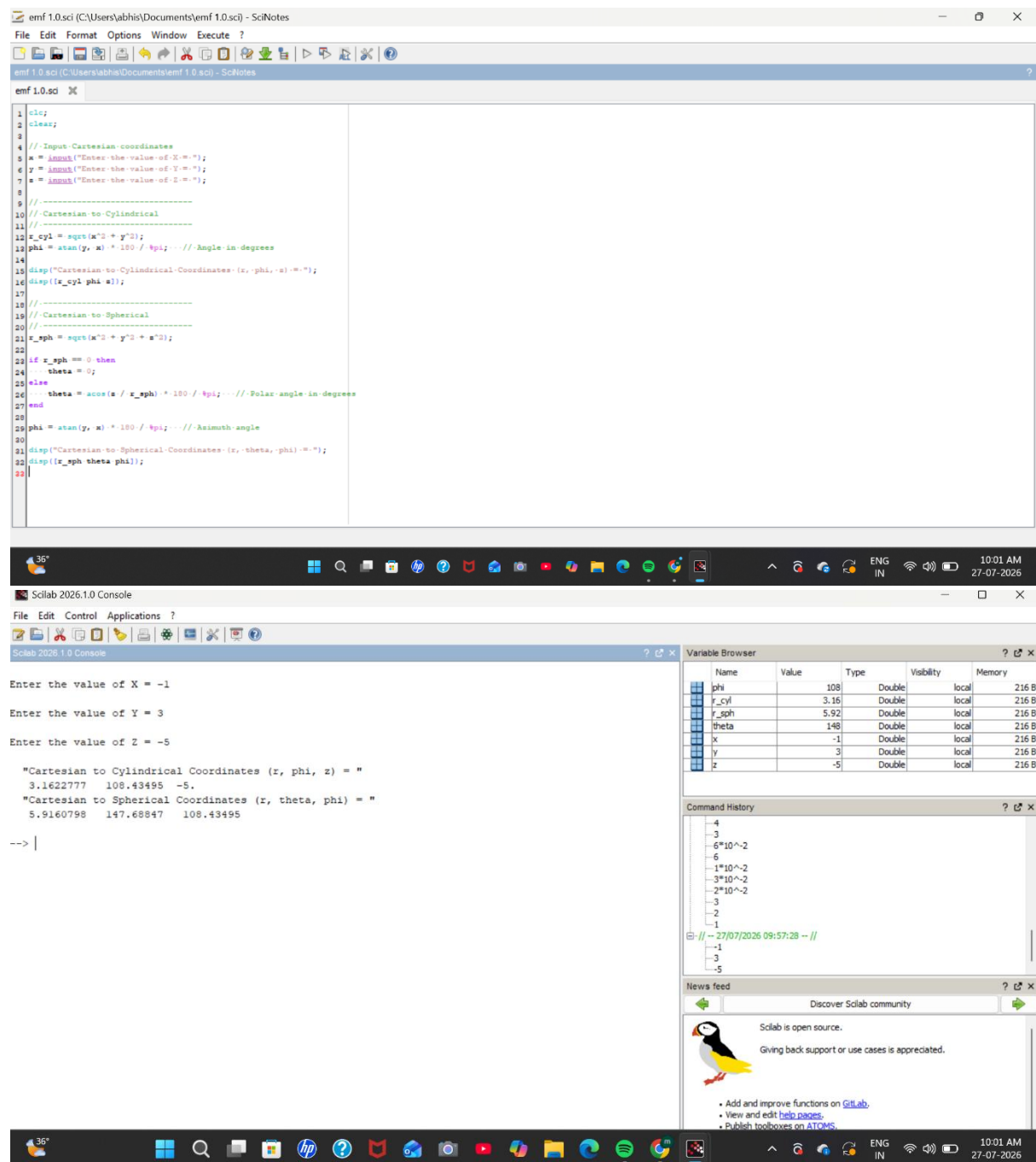
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Emf\_eca0502

Aim:

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system

using SCILAB.



```
1 clear;
2 clear;
3
4 // Input Cartesian coordinates
5 x = input("Enter the value of X :- ");
6 y = input("Enter the value of Y :- ");
7 z = input("Enter the value of Z :- ");
8
9 // Cartesian to Cylindrical
10 //-----
11 r_cyl = sqrt(x^2 + y^2);
12 phi = atan(y, x) * 180 / %pi; // Angle in degree
13 z_cyl = z;
14 disp("Cartesian to Cylindrical Coordinates (x, phi, z) :- ");
15 disp([r_cyl phi z]);
16
17 // Cartesian to Spherical
18 //-----
19 r_sph = sqrt(x^2 + y^2 + z^2);
20 if r_sph == 0 then
21     theta = 0;
22 else
23     theta = acos(z / r_sph) * 180 / %pi; // Polar angle in degree
24 end
25 phi = atan(y, x) * 180 / %pi; // Azimuth angle
26 disp("Cartesian to Spherical Coordinates (r, theta, phi) :- ");
27 disp([r_sph theta phi]);
28
```

Enter the value of X = -1

Enter the value of Y = 3

Enter the value of Z = -5

"Cartesian to Cylindrical Coordinates (x, phi, z) = "

3.1622777 108.43495 -5.

"Cartesian to Spherical Coordinates (r, theta, phi) = "

5.9160798 147.68847 108.43495

--> |

| Name  | Value | Type   | Visibility | Memory |
|-------|-------|--------|------------|--------|
| phi   | 108   | Double | local      | 216 B  |
| r_cyl | 3.16  | Double | local      | 216 B  |
| r_sph | 5.92  | Double | local      | 216 B  |
| theta | 148   | Double | local      | 216 B  |
| x     | -1    | Double | local      | 216 B  |
| y     | 3     | Double | local      | 216 B  |
| z     | -5    | Double | local      | 216 B  |

Command History

```
4
3
6*10^-2
6
1*10^-2
3*10^-2
2*10^-2
3
2
1
// -- 27/07/2026 09:57:28 -- //
```

News feed

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Given

$$q = 3 \times 10^{-6}$$

$$r = 4 \times 10^{-3} \text{ m}$$

Formula

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$r^2 = (4 \times 10^{-3})^2 = 16 \times 10^{-6} = 1.6 \times 10^{-5} \text{ m}^2$$

$$4\pi\epsilon_0 r^2 = (4)(3.1416)(8.854 \times 10^{-12})(1.6 \times 10^{-5})$$
$$= 1.7803 \times 10^{-15}$$

$$E = \frac{3 \times 10^{-6}}{1.7803 \times 10^{-15}} = 1.685 \times 10^9 \text{ V/m}$$

Ans

$$E = 1.576 \times 10^9 \text{ V/m}$$

Electric Flux Density (D)

Formula

$$D = \epsilon_0 E$$

$$D = (8.854 \times 10^{-12})(1.685 \times 10^9)$$

$$(ii) D = 1.492 \times 10^{-2} \text{ C/m}^2$$

$$D = 0.0013951$$

//

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

SciLab 2026.1.0 Console

File Edit Control Applications ?

SciLab 2026.1.0 Console

```

Enter the value of r = 3
Enter the value of phi (degrees) = 30
Enter the value of z = -4

"Cylindrical to Cartesian Coordinates (x, y, z) = "
2.5980762 1.5 -4.
"Cylindrical to Spherical Coordinates (r, theta, phi) = "
5. 143.1301 30.

-->

```

| Name    | Value | Type   | Visibility | Memory |
|---------|-------|--------|------------|--------|
| phi     | 30    | Double | local      | 216 B  |
| phi_sph | 30    | Double | local      | 216 B  |
| r       | 3     | Double | local      | 216 B  |
| r_sph   | 5     | Double | local      | 216 B  |
| theta   | 143   | Double | local      | 216 B  |
| x       | 2.6   | Double | local      | 216 B  |
| y       | 1.5   | Double | local      | 216 B  |
| z       | -4    | Double | local      | 216 B  |

Command History

```

3
2
1
-- 27/07/2026 09:57:28 -- //
-1
3
-5
3
30
-4
dc
3
30
-4

```

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emf 1.1.sci (C:\Users\abhis\Documents\emf 1.1.sci) - SciNotes

File Edit Format Options Window Execute ?

emf 1.1.sci (C:\Users\abhis\Documents\emf 1.1.sci) - SciNotes

emf 1.0.sci emf 1.1.sci

```

1 clear;
2 clear;
3
4 //Input-cylindrical-coordinates
5 r = input("Enter the value of r:-");
6 phi = input("Enter the value of phi (degrees):-");
7 z = input("Enter the value of z:-");
8
9 //-----
10 //Cylindrical-to-Cartesian
11 //-----
12 x = r * cosd(phi);
13 y = r * sind(phi);
14
15 disp("Cylindrical-to-Cartesian-Coordinates-(x,y,z):-");
16 disp([x y z]);
17
18 //-----
19 //Cylindrical-to-Spherical
20 //-----
21 r_sph = sqrt(x^2 + y^2);
22
23 if r_sph == 0 then
24     theta = 0;
25 else
26     theta = asind(x / r_sph);
27 end
28
29 phi_sph = phi;
30
31 disp("Cylindrical-to-Spherical-Coordinates-(r,theta,phi):-");
32 disp([r_sph theta phi_sph]);
33

```

Given Values:

$$\begin{aligned} r &= 3 \\ \phi &= 30^\circ \\ z &= -4 \end{aligned}$$

Formula:

$$\begin{aligned} x &= r \cos \phi \\ y &= r \sin \phi \\ z &= z \end{aligned}$$

Calculation:

$$x = 3 \times \cos(30^\circ)$$

$$x = 3 \times 0.866 = 2.5981$$

$$y = 3 \times \sin(30^\circ)$$

$$y = 3 \times 0.5 = 1.5000$$

$$z = -4$$

$$(x, y, z) = (2.5981, 1.5000, -4)$$

Formula:

$$R = \sqrt{r^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{R}\right)$$

$$\phi = \phi$$

Calculation:

$$R = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\theta = \cos^{-1}\left(\frac{-4}{5}\right) = \cos^{-1}(-0.8) = 143.13^\circ$$

$$\phi = 30^\circ$$

$$(R, \phi, \theta) = (5, 30^\circ, 143.13^\circ)$$

### Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

The screenshot displays the Scilab 2026.1.0 environment. The top window, 'emf 3.2.sci', contains a script for transforming spherical coordinates (r, phi, theta) into Cartesian (x, y, z) and Cylindrical (r\_cyl, phi\_cyl, z\_cyl) coordinates. The script includes input prompts and display commands for the results.

```
1 clear;
2 clc;
3
4 // Input spherical coordinates
5 r = input("Enter the value of r:-");
6 phi = input("Enter the value of phi (degrees):-");
7 theta = input("Enter the value of theta (degrees):-");
8
9 //-----
10 // Spherical to Cartesian
11 //-----
12 x = r * sind(theta) * cos(phi);
13 y = r * sind(theta) * sind(phi);
14 z = r * cosd(theta);
15
16 disp("Spherical to Cartesian Coordinates (x, y, z) = ");
17 disp([x y z]);
18
19 //-----
20 // Spherical to Cylindrical
21 //-----
22 r_cyl = r * sind(theta);
23 phi_cyl = phi;
24 z_cyl = r * cosd(theta);
25
26 disp("Spherical to Cylindrical Coordinates (r, phi, z) = ");
27 disp([r_cyl phi_cyl z_cyl]);
28
```

The bottom window, 'Scilab 2026.1.0 Console', shows the execution of the script. It prompts for the input values: r = 10, phi = 45, and theta = 130. The output displays the Cartesian coordinates (5.4167522, 5.4167522, -6.4278761) and the Cylindrical coordinates (7.6604444, 45, -6.4278761).

The 'Variable Browser' on the right lists the variables and their values:

| Name    | Value | Type   | Visibility | Memory |
|---------|-------|--------|------------|--------|
| phi     | 45    | Double | local      | 216 B  |
| phi_cyl | 45    | Double | local      | 216 B  |
| r       | 10    | Double | local      | 216 B  |
| r_cyl   | 7.66  | Double | local      | 216 B  |
| theta   | 130   | Double | local      | 216 B  |
| x       | 5.42  | Double | local      | 216 B  |
| y       | 5.42  | Double | local      | 216 B  |
| z       | -6.43 | Double | local      | 216 B  |
| z_cyl   | -6.43 | Double | local      | 216 B  |

The 'Command History' panel shows the sequence of commands executed, including the input prompts and the display commands.

The 'News feed' panel at the bottom right contains information about Scilab being open source and provides links to the Scilab community, help pages, and ATOMS toolbox.



### Calculation

$$r = 10 \times 0.7071 = 7.071$$

$$\phi = 130^\circ$$

$$z = 10 \times 0.7071 = 7.071$$

$$(r, \phi, z) = (7.071, 130^\circ, 7.071)$$

$$x, y, z = 7.660444, 4.5, -6.4278761$$

$$r = 10$$

$$\phi = 45^\circ$$

$$\theta = 130^\circ$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\sin 130^\circ = 0.7660444$$

$$\cos 130^\circ = -0.6427876$$

$$\sin 45^\circ = 0.7071068$$

$$\cos 45^\circ = 0.7071068$$

$$x = 10 \times 0.7660444 \times 0.7071068$$

$$x = 5.4167522$$

$$y = 10 \times 0.7660444 \times 0.7071068$$

$$y = 5.4167522$$

$$z = 10 \times (-0.6427876)$$

$$z = -6.4278761$$

$$P(x, y, z) = (5.4167522, 5.4167522, -6.4278761)$$

Formula

$$p = r \sin \theta$$

$$\phi = \phi$$

$$z = r \cos \theta$$

$$p = 10 \times 0.7660444$$

$$p = 7.660444$$

$$z = 10 \times (-0.642787876)$$

$$z = -6.4278781$$

$$P(p, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$$

$$P(x, y, z) = (5.4167522, 5.4167522, -6.4278761)$$

$$P(p, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$$

$$Q = 3 \times 10^6$$

$$r = 4 \text{ mm}$$

$$D = 1.48 \times 10^{-5} \text{ cm}$$

$$D = 0.0012027$$

Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its

classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to

cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.